Instructions:

1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

Q. 1: Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

```
#include<stdio.h>
void func1(int *arr){
  for(int i=1; i<=3;i++)
    (i-1)[arr] = (i%3);
}
void func2 (int a[1] , int arr){
  a[0] = arr + 1;
}
int main(){
  int arr[4] = {0,0,0,0};
  func1(arr+1);
  for(int i=0; i<4; i++)
        printf("%d ", i[arr] );
  printf("\n");
  func2(arr, *arr);
  for(int i=0; i<4; i++)
        printf("%d ", i[arr] );
  return 0;
}
```

Output

0 1 2 0

1 1 2 0

Will the answer change if we write func2 as:

```
void func2 (int a[1] , int arr){
 int temp = arr + 1;
 a = &temp;
}
```

Explain your answer

Yes, ~~because~~ the answer will change because the address of the variable 'temp' will be stored in 'a'. And hence, the array's location will change and it will now ~~start from~~ have a[0] at temp's location

Q. 2: Write the output of the following sequence of instructions **long arr2[100];** **int arr1[100], a1=(&arr2[50]), a2=(&arr2[10]), b1=(&arr1[50]), b2=(&arr1[10]);** **printf("%d %d", a1-a2 , b1-b2);** Explain your answer. (2+2 Marks)

320 160

This is became ~~when~~ the address of arr2[50] - address of arr2[10] will be 320 places apart due to the size of each element of arr2[100] array being 8 bytes (long). Therefore, arr2[50] will be 40×8 = 320 places after arr2[10] inside the memory.

Similarly, due to the size of each element arr1[100] array being 4 bytes (int), arr1[50] will be 40×4 = 160 places after arr1[10] inside the memory.